



Animesh Choudhury

Senior Research Fellow | Remote Sensing & GIS Specialist



PROFESSIONAL SUMMARY

Accomplished researcher specializing in Remote Sensing, GIS, and Climate Science with extensive experience in satellite data analysis, machine learning applications for environmental monitoring, and geospatial modeling. Strong background in drought monitoring, snow cover mapping, air quality analysis, and land-atmosphere interactions. Proficient in Python, R, Google Earth Engine, and various remote sensing tools.

EDUCATION

PGD (Post Graduate Diploma) in Remote Sensing and GIS | 2017-2018

Indian Institute of Remote Sensing (IIRS), India

M.Sc in Environmental Science | 2015-2017

Asutosh College, Calcutta University, India

B.Sc in Physics | 2009-2013

Raghunathpur College, Burdwan University, India

WORK EXPERIENCE

Junior Research Fellow

National Institute of Technology, Rourkela, India | Feb 2022 - Present

- Conducting research on land-atmosphere interaction during thunderstorms over major urban areas of India
- Utilizing high-resolution mesoscale modeling framework and climatological datasets
- Analyzing weather patterns and their impacts on urban environments

Junior Research Fellow

Department of Remote Sensing and GIS, Vidyasagar University, Midnapore, India | Jan 2020 - Feb 2022

- Developed ML models for prediction of agricultural droughts using subseasonal to seasonal forecasting
- Conducted long-term monitoring of droughts using satellite-based indices
- Estimated crop yield reduction using machine learning approaches
- Organized National Workshop on Drought Monitoring and Early Warning using Geospatial Techniques

Junior Research Fellow

DST-Mahamana Centre of Excellence in Climate Change Research, BHU, Varanasi, India | Jun 2019 - Jan 2020

- Analyzed horizontal and vertical distribution of air pollutants over Indo-Gangetic plain
- Utilized ground-based and satellite observations for air quality monitoring
- Presented research on aerosol vertical distribution using CALIPSO LIDAR at AAPW-5

Junior Research Fellow

Indian Institute of Remote Sensing (IIRS), Dehradun, India | Oct 2018 - Mar 2019

- Mapped snow cover area of Northwest Himalaya using MODIS, Sentinel, and Landsat datasets
- Implemented analysis using Google Earth Engine platform
- Installed and supervised automatic weather stations, snow precipitation gauges, and snow pack analyzers
- Contributed to EOAM project on Mountain Ecosystem and Water Resource monitoring

Junior Research Fellow

CSIR - Institute of Himalayan Bioresource Technology, Palampur, India | Aug 2018 - Oct 2018

- Created spectral library of forest tree species found in Himalayan Region
- Collected spectral signatures using ASD Fieldspec spectroradiometer
- Correlated spectral data with bio-physical parameters
- Classified tree species using satellite-based hyperspectral data

SELECTED PUBLICATIONS

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- [1] Budakoti S, Singh C, Choudhury A. **"Transport of a severe dust storm from Middle East to Indian region and its impact on surrounding environment"** *International Journal of Environmental Science and Technology*, 2022. doi:10.1007/s13762-022-04520-1
- [2] Choudhury A, Yadav AC, Bonafoni S. **"A Response of Snow Cover to the Climate in the Northwest Himalaya (NWH) Using Satellite Products"** *Remote Sensing*, 2021. doi:10.3390/rs13040655
- [3] Choudhury A, Dutta D, Bera D, Kundu A. **"Regional variation of drought parameters and long-term trends over India using standardized precipitation evapotranspiration index"** *Journal of Environmental Management*, 2021. doi:10.1016/j.jenvman.2021.113056

CONFERENCE PAPERS & PROCEEDINGS

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- [1] **"Simulation of an Extreme Dust Episode using WRF-CHEM"** *IASTA 2018*, 2018.
- [2] **"Role of Earth Observation Data and Hydrological Modeling in Supporting UN SDGs in North West Himalaya"** *ISPRS Archives*, 2020.
- [3] **"Monitoring of Surface Water of East Kolkata Wetlands (1984-2015) with Global Surface Water Dataset"** *ACRS 2020*, 2020.

TECHNICAL SKILLS

Remote Sensing Tools: ENVI, SNAP, Idrisi, ERDAS Imagine

GIS Software: ArcGIS, QGIS, Google Earth Engine

Programming: Python, R, SQL, MySQL

Data Science & ML: Machine Learning, Deep Learning, Data Analysis, Data Visualization

Domain Expertise: Drought Monitoring, Snow Cover Mapping, Air Quality Analysis, Climate Modeling, Agricultural Remote Sensing

Other Tools: Microsoft Office, WRF-CHEM, CALIPSO LIDAR, MODIS, Sentinel, Landsat

LANGUAGES

Bengali: Native | **English:** Fluent (C2) | **Hindi:** Fluent (C2 Speaking, B2 Reading/Writing)

AWARDS & RECOGNITION

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- **Organizing Member and Speaker**, National Workshop on Drought Monitoring and Early Warning (2021)

PROFESSIONAL DEVELOPMENT

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- Poster presentation at Asian Air Pollution Workshop (AAPW-5), November 2019
 - Attended webinar on Geospatial Techniques in Drought Characterization, September 2020
 - Attended National webinar on Environment and Ecosystem Restoration, June 2021
 - Completed 5-day training on Machine Learning and Deep Learning for Earth Observation, August 2021

Last updated: February 2026